

Total No. of Questions : 8]

[Total No. of Printed Pages : 4

Roll No

EC-222-CBCS

B.E. III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

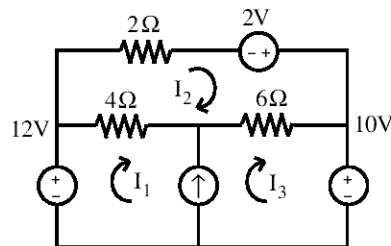
Network Analysis

Time : Three Hours

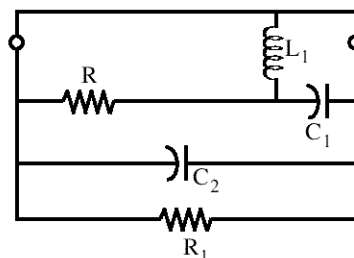
Maximum Marks : 60

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) State KCL and KVL and determine the mesh current I_1 , I_2 and I_3 .



- b) Define graph of the network and draw the graph for the given circuit.

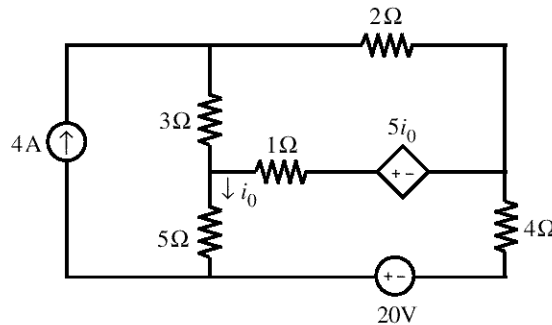


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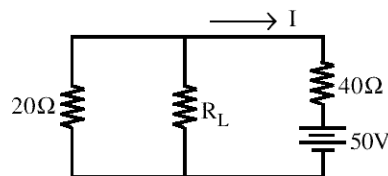
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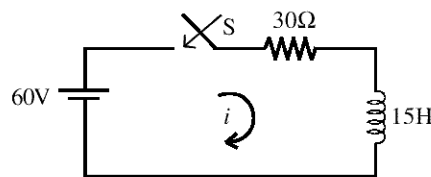
2. a) Find i_0 in the circuit shown using superposition.



- b) In the network of figure, the maximum power is delivered to R_L if its value is



3. a) A series RL circuit with $R = 30\Omega$ and $L = 15H$ has a constant voltage $V = 60V$ applied at $t = 0$ as shown in figure. Determine the current i , the voltage across resistor and the voltage across the inductor.



- b) Explain the transient analysis of series RC circuit.
4. a) Determine the Laplace transform of
- $e^{-at} \cos \omega_0 t u(t)$
 - $t^n e^{-at} u(t)$

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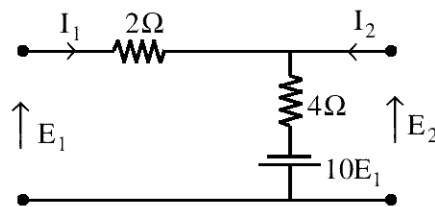
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b) For the $X(s) = \frac{10(2s+3)}{s(s^2+2s+5)}$

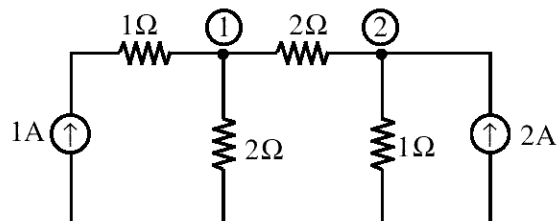
Obtain initial value and final value.

5. a) The z parameters z_{11} and z_{21} for the 2-port networks in figure are:

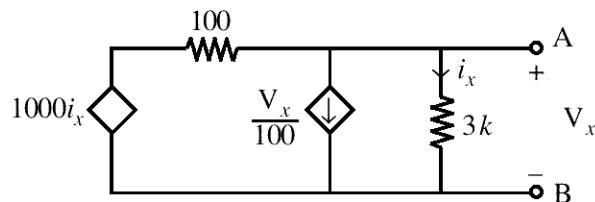


- b) Derive hybrid parameters in terms of z-parameters.

6. a) Find voltage at nodes 1 and 2 from the given figure:



- b) Find the Thevenin and Norton equivalent of the circuit.



[4]

7.
 - a) Explain initial and final value theorem.
 - b) Write down the condition for Reciprocity and condition for symmetry.

8. Write short notes on:
 - a) Incidence matrix and tie set matrices
 - b) Millman's Theorem
 - c) Dot convention and coupling coefficient.
